Detecting Large Language Models in Exam Essays

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***Abstract*—There is a widespread fear that large language models (LLMs) produce content that is indistinguishable from an original human work. Can students use LLM-based tools for their exam essays without being spotted? We performed experiments on a publicly available dataset that we produced, containing thirty answers provided by ChatGPT and thirty-six answers from university students, for each of six anonymized exam essay traces from a scientific methodology course for engineers. We applied term frequency-inverse document frequency (TF-IDF), a state-of-the-art machine learning algorithm for natural language processing, and the cosine similarity metric, in order to produce two LLM detectors. One is based on one-class support vector machine anomaly detection, and the other is based on multi-class random forest classification. The results show that it is possible to spot when LLMs are involved, provided that the source LLM is known. Education institutions can follow our guidelines to prevent cheating and improve education.**

***Index Terms*—large language model, ChatGPT, detector, education, essay, exam**

I. Introduction

Among large language models (LLMs), ChatGPT is a chat interface based on GPT-3, from the class of generative pre-trained transformer (GPT) models. After that OpenAI released ChatGPT, scholars have started testing its abilities and limitations [1]–[3] in any possible fields, from biology [4] to business [5], engineering [6], English language [7], finance [8], [9], journalism [10], law [11], medicine [12]–[17], pharmacology [18], political science [19], programming [20] and psychology [21]. On December 9th 2022, a Nature article titles ”AI bot ChatGPT writes smart essays — should professors worry?” [22]. On December 12th 2022, a daily brief on the same journal [23] asks if ChatGPT will kill the essay assignment or not. Several scholars [6], [24], [25] have presented their perspectives on the future of education with artificial intelligence (AI) tools such as ChatGPT. Susnjak [26] found that ChatGPT presents a significant threat to the integrity of online exams, particularly in the context of university education where online exams are becoming increasingly common, because the model demonstrates a high degree of critical thinking and it is able to generate highly realistic text with little input, making it possible for students to cheat on exams; the author argues that it is crucial for education institutions to research and deploy strategies for combating the risk of cheating with these AI tools. Other scholars, such as King [27], have started using ChatGPT to generate content for scientific articles, even listing ChatGPT as co-author [27]. Another article on Nature [28], dated January 18th, 2023, reports the trend of articles being co-authored by ChatGPT, and presents the responses of several scientific journals and editors who are pro (few) or against (most) such practice. While refraining from such a practice, we found an article written by Cotton et al. [24] - and unofficially by ChatGPT - quite illuminating, because ChatGPT's answers to questions such as ”how can we avoid students cheating to exams with ChatGPT?” with valid advice based on ChatGPT self-knowledge of its answers; however, ChatGPT does not provide an accurate method, e.g., an algorithm to identify its answers, thus we took on the challenge of using AI to spot LLM answers with high precision. Our article aims at answering the following research question (RQ):

RQ: Can we detect LLM answers to known questions with high precision?

This paper is structured as follows. In Section II, we summarize all the related works that we found. In Section III, we describe our research methodology. In Section IV, we report our results, and we discuss them in Section V. Finally, we present our conclusions in Section VI.

II. Related Works

While answering our RQ in January 2023, we looked for similar articles in literature. Guo et al. [29] proposed a human-ChatGPT comparison corpus (HC3) dataset, consisting of nearly 40K questions and their corresponding human-ChatGPT answers. Based on the HC3 dataset, the authors conducted extensive studies including human evaluations, linguistic analysis, and content detection experiments. They provided insights into the implicit differences between humans and LLMs. They also conducted several LLM content detection experiments that can help the research and development of LLM detection tools. While their approach covers a large but finite dataset collected, we focused our research on the detection of answers to known exam traces, characterizing the differences between university students and LLMs answers with a practical and repeatable method. Gao et al. [30] used an AI detector to compare scientific abstracts generated by ChatGPT with real ones. They reached very high accuracy with a GPT-2 output detector [31] based on the RoBERTa [32] sequence classifier; however, ChatGPT is fine-tuned on the GPT-3 model, an upgraded version. Shijaku and Canhasi [33] trained a XGBoost classifier [34] based on the term frequency-inverse document frequency (TF-IDF) method for feature selection that reached high precision, recall and F1-score on both LLM and human generated answers. Their classifier was trained on 252 test of English as a foreign language (TOEFL) essays, 126 humans and 126 non-humans, released in a public dataset. Despite, our works are quite similar, we have experimented our detection system on a dataset generated from an engineering course to which we applied the same state-of-the-art feature extraction but different detection and classification methods. Ventayen [35] focused on comparing the similarity index provided by anti-plagiarism tools. At the time of our study, no other articles on LLM detectors had been published.

III. Research Methodology

In this research, in order to answer our RQ, we compared the answers to six exam traces of an engineering exam given by a LLM-based chat from OpenAI named ChatGPT with the answers given by university students. We took the six exam traces from a 2022 exam session of the [redacted for review] course at [redacted for review]. The six traces are reported in Table I.

TABLE I
THE SIX EXAM TRACES PROPOSED TO STUDENTS AND CHATGPT.

Trace 1	Present and discuss the Gettier objection to the standard definition of knowledge presented in the course. Include your personal opinion.
Trace 2	Discuss the role of the "canons and principles" of ethical behaviour in the context of engineering. How are they different from laws and how are they used?
Trace 3	Describe the different "criteria of truth" and discuss their different application and value though examples.
Trace 4	Reflect on the research activity in the domain of manufacturing by referring to scientific and the engineering design method.
Trace 5	Describe the relationship between science and philosophy by reflecting on the philosophical origin of important scientific activities such as: ask, label, model, decompose, measure, draw and communicate.
Trace 6	Reflect on the following definition of research quality: Q=R*(S+P)

We obtained the LLM-based answers dataset by accessing ChatGPT from OpenAI in its January 9th 2023 version [36]. We queried the LLM-based chat 30 times for each of six exam traces, with a total of 180 queries. In particular, we opened a chat three times and regenerated each answer 10 times; however, we noticed that regenerating an answer instead of producing a new answer in a new chat did not influence the response from ChatGPT, as we obtained completely new answers each time. All the answers from ChatGPT are labelled from "Answer 1" to "Answer 6", corresponding to the six exam traces, from "Chat 1" to "Chat 3", corresponding to the three different chats open for each exam trace, and from "Attempt 1" to "Attempt 10" for each query (Attempt 1) followed by nine regenerated answers (Attempts 2-10). For simplicity, we can count 30 unique answers from our chosen LLM, namely ChatGPT, for each exam trace, with a total of 180 unique short essays. It will be clear from the results section that ChatGPT never gives the same exact answer to the same exact question, neither in a new chat, nor for a regenerated answer.

From the university course, we also took 36 answers for each of the six traces that were given by 36 engineering students, with a total of 216 short essays. The criteria was to select only the students who had given all the six answers in the exam. The students' names were replaced by labels from "Student 1" to "Student 36". The answers were labeled from "Answer 1" to "Answer 6". This exam session dates prior to the release of ChatGPT, thus no students could have used its LLM-based answers to complete their exams.

For each of all the 396 answers, 180 from our chosen LLM, namely ChatGPT, and 216 from students, we created a separate text (.txt) file and stripped it of any special text formats, e.g., tables (used by a few students), or graphics (used by a few students). Then, we imported and pre-processed all the text files with Python, in order to set every character to lower-case and further remove any end-line-characters, punctuation, or special characters, with the exception of the [a-z] and [0-9] characters.

We applied a natural language processing (NLP) machine learning (ML) algorithm called TF-IDF to vectorize all the answers. We used the function TfidfVectorizer() in feature_extraction.text from the scikit-learn Python library. After the vectorization, we applied a cosine similarity (CS) metric to compare the 30 vectorized unique short essays from the LLM with the students' 36 vectorized short essays, for each of the six exam traces.

The cosine similarity (see Equation 1) is a scalar value in the range [0,1], where a value of 0 indicates completely different text vectors and a value of 1 indicates completely similar text vectors, calculated with the following formula:

$$CS = \cos(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A}\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} \mathbf{A}_i \mathbf{B}_i}{\sqrt{\sum_{i=1}^{n} (\mathbf{A}_i)^2} \sqrt{\sum_{i=1}^{n} (\mathbf{B}_i)^2}}, \tag{1}$$

where $\mathbf{A}$ and $\mathbf{B}$ are two text vectors, i.e., a pair of vectorized short essays being compared.

Each of the six answers comparison led to a cosine similarity heatmap reported in Section IV-A.

We further analyzed how LLMs and students' answers appear in an embedding space [37]. In particular, we applied a pre-trained transformer model to calculate the answer embeddings, fine tuned from Microsoft/MPNet-base. A tokenizer processes the words according to a vocabulary, defined in the phase of training or fine-tuning, which maps each word with the relative token id. The model receives the tokens of a phrase and gives in output an embedding vector of shape [1,768], independently form the input data length. We used a t-distributed stochastic neighbor embedding (TSNE) reduce the dimensionality of the embedding spaces from 768 features to two. The results are reported in Section IV-B.

After the data analysis was completed, we built two types of LLM detectors: for the first type of detector, we used a one-class dataset, i.e. only LLMs answers; for the second type of detector, we used a multi-class dataset, i.e. students' exams and LLMs answers.

The first type of detector can be characterized as an anomaly detector. We can call it a one-class LLM detector. The ML algorithm that we chose is one-class support vector machine (1-SVM) with radial basis function kernel. The reason behind this choice is that 1-SVM remain the state-of-the-art in outlier detection, while other algorithms perform less efficiently, e.g., we tested local outlier factor and minimum covariance determinant, which resulted in markedly worse performances compared to 1-SVM. The training set for 1-SVM consisted of 25 of the 30 LLMs answers for each of the exam traces. We used the other five LLMs answers and the 36 students' answers as our test set. We vectorized both the training and test sets with TF-IDF. The training set was used as a corpus for TF-IDF and the test set was fitted to it. We trained six 1-SVMs, one for each of the exam traces.

The second type of detector can be characterized as a classifier. We can call it a multi-class LLM detector. The ML algorithm that we chose is random forest (RF). Given the simplicity of the classification task (see the results in Table V) when the dataset was composed of separable examples from two labelled and well-balanced classes (in our case, 30 and 36 samples, respectively from LLMs and students, for each exam trace), we did not introduce other classifiers for comparison, as anyone we picked had similarly outstanding performances in our case, e.g., we tested a naïve Bayes classifier and a c-support vector classifier. The training set for RF consisted of 30 of the 36 students' answers and 25 of the 30 LLMs answers for each of the exam traces. We used the other six students' answers and the other five LLMs answers as our test set. We vectorized both the training and test sets with TF-IDF. The training set was used as a corpus for TF-IDF and the test set was fitted to it. We trained six RFs, one for each of the exam traces.

We evaluated the performance of our LLM detectors with the following metrics: precision, recall, F1-score and Matthews correlation coefficient (MCC).

We report our LLM detection results in the next section and further analyze them in sections Discussion and Conclusions of this article.

The dataset composed of the 396 answers that we collected and the Python code that we used to produce our results are publicly available for download on our GitHub repository for this project [redacted for blind review]. Further details of the Python libraries that we used, and all the installation requirements are in our repository.

In Table II, we propose two use cases with the necessary steps to use our two LLM detectors. Use case 1 makes use of our one-class LLM detector and use case 2 makes use of our multi-class LLM detector. These two use cases can be even combined together, as we suggest with an example provided in Table II.

TABLE II
USE CASES FOR OUR LLM DETECTORS.

Use case 1	
Goal	Collect and use only a few LLMs answers to the traces for the current exam session, in order to create a one-class LLM detector.
Step 1	For each exam trace, collect at least 20 answers from a LLM.*
Step 2	Collect all the students' answers from the current exam.*
Step 3	Download and run the python code for the one-class LLM-Detector available in our linked repository.
Use case 2	
Goal	Collect and use a few answers from LLM and a few students' answers from previous exam sessions that can be re-proposed in the current exam session, in order to create a multi-class LLM classifier.
Step 1	For each exam trace, collect at least 20 answers from a LLM.*
Step 2	For each exam trace, collect at least 20 students' answers from past exams.*
Step 3	Collect all the students' answers from the current exam.*
Step 4	Download and run the python code for the multi-class LLM-Detector available in our linked repository.
Use cases 1 & 2 combined	
Goal	Apply case 1 and 2 separately on different exam traces.
Example	Combine four new traces (use case 1) with two traces from previous exams (use case 2). For this application, use both the classifiers available in our repository, as indicated in use cases 1 and 2.

*We suggest to save each answer in a separate file and use category-labelled and numbered file names.

IV. RESULTS

This section is divided in two parts. Firstly, we analyze the cosine similarity of the data we collected. Secondly, we analyze the performance of our LLM detectors.

A. Cosine Similarity Analysis

In Figure 1, we report the cosine similarity heatmaps, between vectorized short essays from students and LLMs, as answers for exam traces from one to six. In Table III, we report the cosine similarity average, minimum and maximum values, comparing the same vectorized short essays as answers to the aforementioned exam traces.

The comparison of cosine similarities for the short essays answering each of the six exam traces (see Figure 1) shows the two aforementioned clusters between students' answers and LLMs answers. We can confirm what we see from the heatmaps in Figure 1 by looking at the values in Table III.

We can see that the highest cosine similarity is among LLM-LLM comparisons, with average values of, respectively, 0.54, 0.65, 0.60, 0.56, 0.52 and 0.58, for answers from 1 to 6.

The second highest cosine similarity is among student-student comparisons, with average values of, respectively, 0.46, 0.45, 0.42, 0.48, 0.45 and 0.51, for answers from 1 to 6.

The lowest cosine similarity is among hybrid student-LLM comparisons, with average values of, respectively, 0.40, 0.43, 0.38, 0.36, 0.38 and 0.43, for answers from 1 to 6.

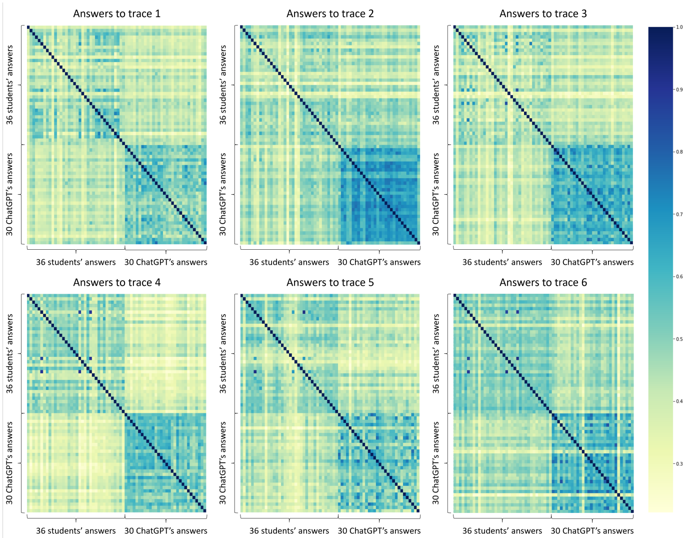

Fig. 1. Cosine similarity heatmaps between vectorized short essays for exam traces one to six. In blue the high CS values and in yellow the low CS values. All the heatmaps show two clear clusters: one for the 36 students' essays and one for the 30 LLMs essays. They also show a lower cosine similarity between students' essays and LLMs essays, an essential feature that an algorithm can exploit to learn to recognize the two classes.

TABLE III
COSINE SIMILARITY AVERAGE, MINIMUM AND MAXIMUM VALUES FOR THE ANSWERS TO THE SIX EXAM TRACES FROM STUDENTS AND LLMS.

Cosine similarity				
		Average	**Min**	**Max**
Answer 1	**student-student**	0.46	0.26	0.74
	LLM-LLM	0.54	0.36	0.78
	student-LLM	0.40	0.22	0.56
Answer 2	**student-student**	0.45	0.20	0.65
	LLM-LLM	0.65	0.38	0.80
	student-LLM	0.43	0.22	0.59
Answer 3	**student-student**	0.42	0.20	0.83
	LLM-LLM	0.60	0.42	0.78
	student-LLM	0.38	0.19	0.55
Answer 4	**student-student**	0.48	0.29	0.90
	LLM-LLM	0.56	0.38	0.70
	student-LLM	0.36	0.22	0.55
Answer 5	**student-student**	0.45	0.20	0.93
	LLM-LLM	0.52	0.26	0.76
	student-LLM	0.38	0.18	0.57
Answer 6	**student-student**	0.51	0.31	0.93
	LLM-LLM	0.58	0.21	0.83
	student-LLM	0.43	0.21	0.60

These three distinct levels of cosine similarities indicate that all the different answers from LLM share enough similarities from one to another, and at the same time enough differences with the students' answers, to allow us to distinguish between the two clusters, i.e. LLMs answers and students' answers.

B. Embedding Analysis

With the embedding analysis (see Figure 2), we show how LLMs and students' answers appear in an embedding space. The TSNE reduction correctly collocates the different traces in the reduced space, showing six clusters, one for each exam trace. Furthermore, we colored each embedding by its label, and we can see that LLMs answers (orange in Figure 2) and students' answers (blue in Figure 2) appear to have a clear position in the two-dimensional embedding space, forming separable sub-clusters (identified by their colors). The shape of the sub-clusters suggests using a radial basis function to separate them, which is what we have done with 1-SVM. The results are reported in the next section.

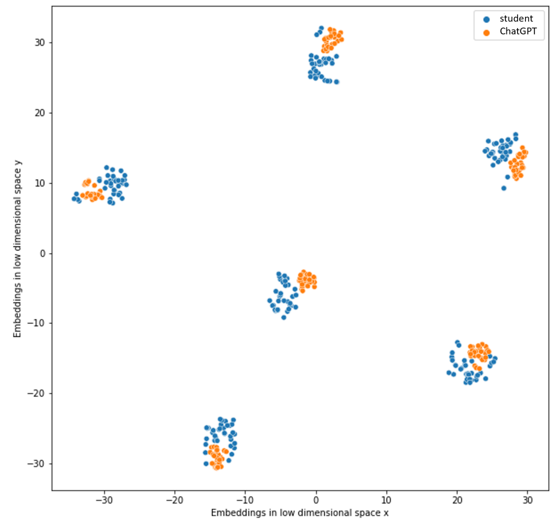

Fig. 2. Two-dimensional (x and y vectors) embedding space obtained from TSNE. Each cluster represents the answers to one of the six exam traces. The embeddings for student's answers and LLMs answers (respectively, blue and orange dots) result separable, e.g., with a radial basis function.

C. One-Class LLM Detector

The one-class LLM detector built with 1-SVM is capable of distinguishing between the LLMs answers that we used to train it, and any other answers that are considered as anomalies or outliers. The students' answers are considered anomalies as they are not seen during training.

We calculated the precision, the recall, the F1-score and the MCC of this one-class LLM detector for each of the answers from students or LLM. All the values are reported in Table IV.

Since precision, recall and F1-score are all affected by imbalanced data [38], which is our case because the test set includes 36 students' answers and only five LLMs answers to detect (see test set in Table IV), the MCC constitutes the best metric to use. The values of MCC in Table IV are the best metrics to evaluate the overall performance of our 1-SVM LLM detectors for the six exam traces. Additionally to these metrics, we report the confusion matrices (CM) of this one-class LLM detector in Figure 3.

D. Multi-Class LLM Detector

The multi-class LLM detector built with RF is capable of distinguishing between both the LLMs answers and the students' answers that we used to train it, i.e., it is a proper classifier.

TABLE IV
RESULTS FOR THE ONE-CLASS LLM DETECTOR WITH 1-SVM FOR EACH TRACE.

Trace	Answer	Train Set	Test Set	Prec.	Recall	F1 Score	MCC
1	student LLM	- 25	36 5	0.93 0.27	0.78 0.60	0.85 0.37	0.28
2	student LLM	- 25	36 5	0.95 1.00	1.00 0.60	0.97 0.75	0.75
3	student LLM	- 25	36 5	0.95 1.00	1.00 0.60	0.97 0.75	0.75
4	student LLM	- 25	36 5	0.92 1.00	1.00 0.40	0.96 0.57	0.61
5	student LLM	- 25	36 5	0.92 0.40	0.92 0.40	0.92 0.40	0.32
6	student LLM	- 25	36 5	0.97 0.80	0.97 0.80	0.97 0.80	0.77

TABLE V
RESULTS FOR THE MULTI-CLASS LLM DETECTOR WITH RF FOR EACH TRACE.

Trace	Answer	Train Set	Test Set	Prec.	Recall	F1 Score	MCC
1	student LLM	25 25	11 5	1.00 1.00	1.00 1.00	1.00 1.00	1.00
2	student LLM	25 25	11 5	1.00 1.00	1.00 1.00	1.00 1.00	1.00
3	student LLM	25 25	11 5	1.00 1.00	1.00 1.00	1.00 1.00	1.00
4	student LLM	25 25	11 5	1.00 1.00	1.00 1.00	1.00 1.00	1.00
5	student LLM	25 25	11 5	1.00 1.00	1.00 1.00	1.00 1.00	1.00
6	student LLM	25 25	11 5	1.00 1.00	1.00 1.00	1.00 1.00	1.00

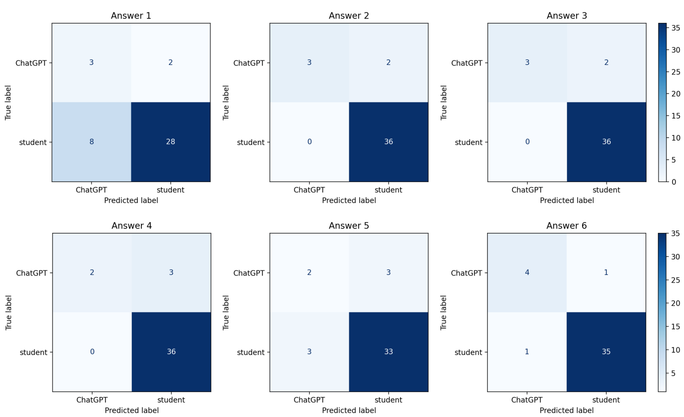

Fig. 3. Confusion matrices for the one-class LLM detector with 1-SVM. Each matrix (one for each exam trace) shows a great ability of 1-SVM to recognize a student's essay as a student's essay, i.e. the anomaly detection. The algorithm can fairly recognize the writing of LLM in answers 2, 3, 4 and 6, but it tends to confuse a few students for LLM in answers 1 and 5. This is reflected into the MCC values in Table IV.

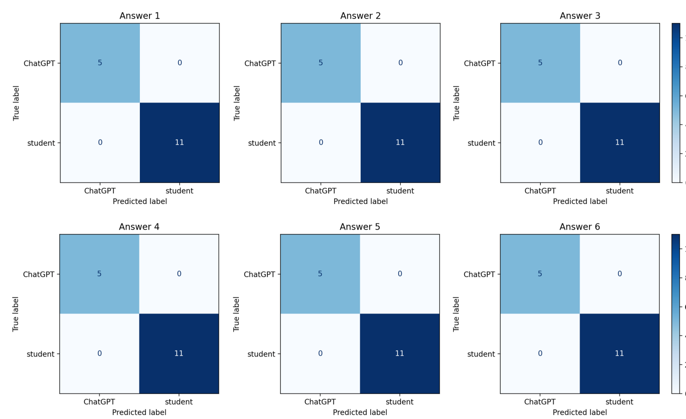

Fig. 4. Confusion matrices for the multi-class LLM detector with RF. Each matrix (one for each exam trace) shows a great ability of RF to recognize a student's essay as a student's essay, i.e. the anomaly detection. The algorithm can recognize with 100% of precision, recall, F1-score and MCC the writing of LLM in all the answers to all traces. This is reflected into the values in Table V.

We calculated the precision, the recall, the F1-score and the MCC of this multi-class LLM detector for each of the answers from students or LLM. All the values are reported in Table V. We also report the confusion matrices of this multi-class LLM detector in Figure 4.

The MCC constitutes the best metric in case of class imbalanced data, which is still our case because the test set includes 11 students' answers and five LLMs answers to classify (see test set in Table V). The imbalance is not as strong as in the previous case (the one-class LLM detector) but the values of MCC in Table V are nonetheless the best metrics to evaluate the overall performance of our RF multi-class LLM detectors for the six exam traces.

V. DISCUSSION

The comparison of cosine similarities (see Section IV-A) for all the six answers to the corresponding six exam traces proposed to students and LLM shows two clear clusters: one from the students' answers and one for the LLMs answers. In almost all cases, the LLM cluster is better defined by a high level of similarity among answers. There is always low cosine similarity between students and LLMs answers. These results indicate us that it is generally possible to build a classifier and detect the AI-based answers.

With the support of the CS analysis, we have built two types of LLM detectors. Our first attempt was to consider a real use case, in which a professor knows the exam traces that they are going to use for their students, and before the exam, they generate a series of answers for those traces from a LLM. A one-class classifier or anomaly detector can thus be trained. We have shown in Section IV-C that such a detector can perform well (see the MCC values in Table IV), but not perfectly, as it is affected by the high class imbalance between the multiple positive samples from the LLM detector and the total absence of negative samples from the students.

We improved the LLM detection to a precision, recall, F1-score and MCC of 100% (see Table V), thus answering our RQ (see Section I), by considering a good multi-class classifier, able to perform with multiple features after the TF-IDF vectorization of the two classes of answers, i.e. from a LLM and the students. Our choice of classifier was for RF. We imagined that in a real scenario, the same professor could have acquired some of the answers of her students from previous exams, and re-proposed a few old traces to her new students. By doing this, the dataset is enriched with a balanced amount of data: the positive examples from a LLM and the negative examples from the old students answers. Considering that the course program, and the teacher usually do not change, this scenario is plausible. The multi-class LLM detector can determine with 100% precision, recall, F1-score or MCC if a student cheated or not.

The two LLM detectors that we proposed are both valid, and have to be considered based on the application required and availability of data. In both cases, we assumed that the AI model generating answers was known, i.e., we have used LLMs. Future works have to address the case in which answers might come from multiple LLMs. We also have imagined that the students would not repeatedly use a LLM to produce small modules of their answers by asking several questions and then collating the answers. This would make the identification of hybrid student-LLM answers more complex. Future studies should address this case.

VI. Conclusions

We proposed two LLM detectors to address the academic fear that students might use LLMs to cheat on their exams, and we showed that LLMs cannot escape them. We built these detectors based on different use cases and we provided a dataset and the code to reproduce our results. This research can be already applied during students' homework or exams, however, future work should focus on overcoming the limitations of the experimental setup we made, and also expand the dataset used for validation. Hopefully, humans will be able to continue outsmarting LLM-empowered cheaters.

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